

Production Ready Web Standard (PRWS)

Rule Engine Specification

Version: 1.0

Status: Draft

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1. Purpose

This document defines the rules, scoring methodology, validation procedures, severity levels, and certification criteria used by the Production Ready Web Standard (PRWS) audit engine.

The purpose of the rule engine is to provide a consistent and objective method of evaluating website readiness for production deployment.

2. Rule Engine Overview

The PRWS Rule Engine shall:

1. Crawl a target website.
 2. Collect metadata.
 3. Analyze technical configurations.
 4. Evaluate compliance against PRWS rules.
 5. Generate findings.
 6. Calculate category scores.
 7. Calculate overall readiness score.
 8. Determine certification eligibility.
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3. Audit Workflow

Step 1

User submits URL

Example:

https://example.com

↓

Step 2

Crawler fetches:

- Homepage
- Linked legal pages
- robots.txt
- sitemap.xml

↓

Step 3

Analyzer evaluates rules

↓

Step 4

Score calculated

↓

Step 5

Report generated

↓

Step 6

Certification determined

4. Severity Levels

Each finding shall be assigned a severity level.

Critical

A missing item that creates legal, security, or operational risk.

Examples:

- No HTTPS
- Invalid SSL
- Missing Privacy Policy
- Exposed Secrets

Penalty:

100% rule failure

High

Significant issue affecting trust or functionality.

Examples:

- Missing Terms of Service
- Missing Security Headers
- Broken Sitemap

Penalty:

75% rule failure

Medium

Issue affecting quality or user experience.

Examples:

- Missing Open Graph Tags
- Missing Contact Page

Penalty:

50% rule failure

Low

Minor issue or recommendation.

Examples:

- Missing Favicon
- Missing Canonical URL

Penalty:

25% rule failure

5. Category Weights

Security

Weight: 30%

Maximum Points: 300

Legal

Weight: 15%

Maximum Points: 150

Reliability

Weight: 15%

Maximum Points: 150

Accessibility

Weight: 15%

Maximum Points: 150

Authentication

Weight: 10%

Maximum Points: 100

SEO

Weight: 10%

Maximum Points: 100

AI Readiness

Weight: 5%

Maximum Points: 50

Total Points:

1000

6. Legal Rules

LEG-001

Rule Name:

Privacy Policy Present

Requirement:

Website must contain a privacy policy.

Detection:

Search for:

/privacy

/privacy-policy

Links containing:

Privacy Policy

Points:

50

Severity:

Critical

LEG-002

Terms of Service Present

Detection:

Search for:

/terms

/terms-of-service

Points:

50

Severity:

High

LEG-003

Cookie Policy

Detection:

Cookie policy page exists.

Points:

25

Severity:

Medium

LEG-004

Copyright Information

Detection:

Footer contains copyright statement.

Points:

25

Severity:

Low

7. Contact Rules

CON-001

Contact Page Exists

Detection:

/contact

Contact navigation link

Points:

25

Severity:

Medium

CON-002

Support Email Exists

Detection:

mailto links

Visible support address

Points:

25

Severity:

Medium

CON-003

Organization Information

Detection:

About page exists.

Points:

25

Severity:

Low

8. Security Rules

SEC-001

HTTPS Enabled

Detection:

SSL certificate valid

Points:

100

Severity:

Critical

SEC-002

HSTS Enabled

Detection:

Strict-Transport-Security header

Points:

30

Severity:

High

SEC-003

Content Security Policy

Detection:

Content-Security-Policy header

Points:

30

Severity:

High

SEC-004

X-Frame-Options

Detection:

Header present

Points:

20

Severity:

Medium

SEC-005

X-Content-Type-Options

Detection:

Header present

Points:

20

Severity:

Medium

SEC-006

Referrer Policy

Detection:

Header present

Points:

20

Severity:

Medium

SEC-007

No Exposed Environment Variables

Detection:

Scan HTML source.

Check for:

API Keys

Secret Tokens

Database Credentials

Points:

80

Severity:

Critical

9. Reliability Rules

REL-001

Custom 404 Page

Detection:

Request invalid route.

Verify custom response.

Points:

40

Severity:

Medium

REL-002

No Broken Internal Links

Detection:

Link crawler.

Points:

50

Severity:

High

REL-003

Monitoring Indicator

Detection:

Known monitoring scripts.

Points:

30

Severity:

Low

REL-004

Favicon Exists

Detection:

favicon.ico

Points:

10

Severity:

Low

10. SEO Rules

SEO-001

robots.txt

Detection:

/robots.txt

Points:

20

Severity:

Medium

SEO-002

sitemap.xml

Detection:

/sitemap.xml

Points:

20

Severity:

Medium

SEO-003

Title Tag

Detection:

HTML title exists

Points:

10

Severity:

Low

SEO-004

Meta Description

Detection:

Meta description exists

Points:

10

Severity:

Low

SEO-005

Open Graph Tags

Detection:

og:title

og:image

og:description

Points:

20

Severity:

Medium

SEO-006

Canonical URLs

Detection:

rel="canonical"

Points:

20

Severity:

Low

11. Accessibility Rules

ACC-001

Image Alt Attributes

Detection:

Images contain alt text

Points:

40

Severity:

Medium

ACC-002

Form Labels

Detection:

Forms linked to labels

Points:

40

Severity:

Medium

ACC-003

Semantic HTML

Detection:

header

main

footer

nav

Points:

30

Severity:

Low

ACC-004

Language Declaration

Detection:

html lang=""

Points:

20

Severity:

Low

12. Authentication Rules

AUTH-001

Secure Cookies

Detection:

Secure flag

Points:

30

Severity:

High

AUTH-002

HttpOnly Cookies

Detection:

HttpOnly flag

Points:

30

Severity:

High

AUTH-003

Password Reset Mechanism

Detection:

Reset workflow exists

Points:

20

Severity:

Medium

AUTH-004

Email Verification

Detection:

Verification workflow exists

Points:

20

Severity:

Medium

13. AI Readiness Rules

AI-001

AI Disclosure

Requirement:

Website discloses AI-generated content or AI features.

Points:

10

Severity:

Low

AI-002

Rate Limiting

Detection:

API protection mechanisms.

Points:

20

Severity:

High

AI-003

Prompt Injection Protection Notice

Detection:

AI safety policy exists.

Points:

10

Severity:

Medium

AI-004

Model Usage Terms

Detection:

AI terms page exists.

Points:

10

Severity:

Medium

14. Scoring Formula

Category Score

Formula:

$\text{Earned Points} / \text{Available Points} \times 100$

Example:

$240 / 300 \times 100$

$= 80\%$

Overall Score

Formula:

Weighted Category Sum

Example:

Security = 80%

Legal = 100%

SEO = 70%

Accessibility = 90%

Final Score:

85%

15. Certification Rules

Bronze

Requirements:

Score ≥ 60

No Critical Failures

Silver

Requirements:

Score ≥ 75

No Critical Failures

Gold

Requirements:

Score ≥ 90

Maximum 2 High Findings

Platinum

Requirements:

Score ≥ 95

No Critical Findings

No High Findings

16. Rule Engine Extensibility

All rules shall be stored as configuration records.

Example:

```
{ "rule_id": "SEC-001", "name": "HTTPS Enabled", "category": "Security", "points": 100, "severity": "Critical" }
```

This allows rules to be updated without changing application code.

17. Versioning Policy

PRWS shall use semantic versioning.

Examples:

PRWS 1.0.0

PRWS 1.1.0

PRWS 2.0.0

Major Version

Breaking rule changes

Minor Version

New rules added

Patch Version

Clarifications and bug fixes

END OF SPECIFICATION